

16,000

THE NUMBER OF **WORKERS**
ASSEMBLING **SAILING SHIPS**
AT THE **VENICE ARSENALE**
IN THE 17TH CENTURY



A 17th-century painting shows workers at Venice's Arsenale. Innovative construction techniques enabled the Venetians to dominate the seaways for centuries.

AROUND 1104, the city authorities in Venice ordered the construction of the Arsenale, a state shipyard and armory, which would employ 16,000 workers by the 17th century. The Arsenale **pioneered new production techniques**, producing prefabricated parts and

the end of the Han Dynasty (220 CE), but they began to use **multiple colors** in the wooden block printing of pictures c.1107. By 1340, the technique was applied to an edition of the *Diamond Sutra* (see 861–99), in which the



Raymond of Toledo
Archbishop Raymond is seen standing before King Alfonso VII at his coronation in 1135, a demonstration of the importance of royal patronage.

Around 1121, in the Persian city of Merv, **al-Khazini** wrote *Book of the Balance of Wisdom* in which he put forward a **theory of centers of gravity**. He suggested gravity varies according to the distance from the center of the world—the farther the objects are, the heavier they seem.

English philosopher **Adelard of Bath** (1080–1152) spent seven years in Salerno and Sicily, where he learned Arabic. His extensive knowledge of Arabic culture and language led him in 1126 to **translate al-Khwarizmi's astronomical work**, the *Sindhind Zij*, (*Astronomical Tables of Sindhind*) into Latin, bringing his work to a wider audience.

Raymond, Archbishop of Toledo in Spain (1126–52), **encouraged the translation of books from Arabic into Latin**. The first translators were succeeded in 1167 by **Gerard of Cremona** (1114–87), who translated more than 80 Arabic works.

“BECAUSE OF THE **FREQUENCY** OF THE **EXPERIENCE**, THESE JUDGMENTS MAY BE REGARDED AS CERTAIN, EVEN WITHOUT OUR **KNOWING THE REASON.**”

Abu'l Barakat al-Baghdadi, in *Kitab al-Mu'tabar*, early 12th century

using a method of frame-building for ships that made it possible to **construct a vessel in a day**.

The Chinese had started to print on silk using stencils before

main text is in black, and the prayers are in red.

In the 11th century, Avicenna had theorized that the **motion of a projectile** continues

because of the *mail* (inclination, or motive power) imparted to it by the projector, but said that only one such force could exist in a body at any time. This was later confirmed by French priest Jean Buridan (see 1350–62). Around 1120, Baghdad philosopher **Abu'l Barakat** (c.1080–1165) suggested that more than one *mail* could exist in a projectile. As it fell, the *mail* pushing it forward weakened, and another *mail* took over, causing it to accelerate downward. These *mail* forces caused acceleration. In this way, he expressed the idea of the relationship between force and acceleration.

TRANSLATING ANCIENT MANUSCRIPTS

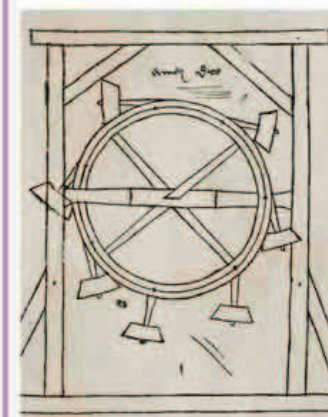
The works of many classical philosophers had been lost in the Christian West, but they were preserved through translations made into Arabic in the 8th and 9th centuries. These manuscripts in turn became available in Europe from the 12th century, where they were translated into Latin by scholars such as Gerard of Cremona.



INDIAN MATHEMATICIAN AND ASTRONOMER Bhaskara II (1114–85) described a **perpetual motion machine**, one that would, once a force was imparted to it, continue to work indefinitely.

Bhaskara's device was a **wheel whose spokes were filled with mercury**. He theorized that mercury was sufficiently heavy so that as the wheels turned it would flow to the edge of the spokes and impel the machine around another part-turn.

Bhaskara II was better known for his **astronomical and mathematical works**, which made him one of the most respected Indian mathematicians of the Middle Ages. In *Lilavati* (named after his daughter), his



In perpetual motion
This 13th century version of a perpetual motion machine is an overbalanced wheel with hinged mallets around its rim.

1104 Construction of the Venice Arsenale begins

c.1107 Chinese use multiple colors in woodblock printing

1120 Abu'l Barakat expresses notion of the relationship between force and acceleration

1121 Persian scholar al-Khazini proposes an early form of theory of gravitation

1125 Chinese fleet uses magnetic floating compass on voyage to Korea

1126–51 Raymond of Toledo orders translations of many classical works from Arabic into Latin

1126 Adelard of Bath translates Euclid's *Elements* from Arabic into Latin

1145 Spanish Jewish mathematician Savasorda produces work on quadratic equations

1150 Bhaskara II demonstrates that a number has two square roots, one positive and the other negative

1150 Trotula, one of Europe's first known female physicians, practices at Salerno, Italy